Jacob A. Johnson

Madison, WI | jajohnson66@wisc.edu | github.com/jajohnsonstats

EDUCATION

Ph.D. in Statistics, *University of Wisconsin-Madison* Expected May 2029

B.S. in Statistical Science, Minor in Mathematics, *Brigham Young University* June 2024

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Statistics Intern; *Sandia National Laboratories* May 2025 – Present

- Developed statistical machine learning models to forecast extreme temperatures at subseasonal (2–8 weeks out) timescales across the contiguous United States.
- Evaluated and developed various dimension reduction techniques to extract predictive features from 40+ years of atmospheric data to be used as inputs for temperatures forecasting.

Financial Data Analytics Intern; *Williams International* May 2024 – Aug 2024

- Created statistical models to forecast inventory using historical production and usage data.
- Developed an anomaly detection program to monitor and identify deviations in expenses, detecting unusual spending patterns and potential fraud across business units.
- Automated forecasting and anomaly detection processes, reducing manual oversight.

Research Assistant; *Brigham Young University* May 2022 – April 2024

- Utilized Bayesian machine learning techniques to combine climate data products for predicting average monthly precipitation in the Himalayan Mountains.
- Developed new subsampling methods in R to speed up computation of Gaussian Process calculations for large spatial data.

PUBLICATIONS

[1] Matthew J. Heaton and **Jacob A. Johnson**. "Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes". In: *Bayesian Analysis* (2024), pp. 1–23. DOI: 10.1214/24-BA1459.

[2] **Jacob A. Johnson**, Matthew J. Heaton, William F. Christensen, Lynsie R. Warr, and Summer B. Rupper. "Fusing Climate Data Products Using a Spatially Varying Autoencoder". In: *Journal of Agricultural, Biological and Environmental Statistics* (Oct. 2024). DOI: 10.1007/s13253-024-00657-3.

AWARDS & HONORS

- UW-Madison Statistics Department Outstanding Performance as a TA award 2024 – 2025
- ASA Section on Statistics and the Environment Student Paper Competition Winner 2024